True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. **Degree of precision.** The degree of precision of the following quadrature rule is 3:

$$\int_0^1 f(x)\,\mathrm{d}x \approx \frac{f(0)+f(1)}{2} - \frac{f'(1)-f'(0)}{12}\,.$$

2. **Degree of precision.** The degree of precision of Simpson's 3/8 rule,

$$\int_0^3 f(x)\,\mathrm{d}x \approx \frac{3}{8}\Big(f(0)+3f(1)+3f(2)+f(3)\Big),$$

is equal to 4.

3. **Degree of precision.** If the quadrature rule

$$\int_{-1}^1 f(x)\,\mathrm{d}x \approx \sum_{i=1}^n w_i f(x_i)$$

has degree of precision at least 0, then $w_1 + \cdots + w_n = 2$.

4. **Composite quadrature.** Suppose that $f \in C^\infty[a,b]$ and let $I_n[f]$ denote the approximation of the integral $I[f] \coloneqq \int_a^b f(x)\,\mathrm{d}x$ obtained with the composite midpoint rule with n cells, that is

$$I_n[f] = h\sum_{i=0}^{n-1} f\left(x_i + \frac{h}{2}\right), \qquad h = \frac{b-a}{n}, \qquad x_i = a + ih\,.$$

Then there exists a constant $C \in (0,\infty)$ such that

$$\forall n \in \{1,2,\dots\}, \qquad \Big|I[f] - I_n[f]\Big| \leqslant \frac{C}{n^2}\,.$$

Hint: Compute the degree of precision of the midpoint rule $\int_{-1}^1 f(x)\,\mathrm{d}x \approx 2f(0)$, and use the theorem relating the degree of precision of a quadrature rule to the order of convergence, with respect to h, of the associated composite rule.

5. **Composite quadrature.** With the notation of the previous item, it holds that $I_n[f] = I[f]$ for all $n \geqslant 1$ whenever f is a polynomial of degree at most 2.

6. **Composite quadrature.** Let $f \in C^\infty[a,b]$. For large n, doubling the number of cells in the composite Simpson rule reduces the error by approximately a factor 8.

7. **Composite quadrature.** Let $f\colon [a,b] \to \mathbb{R}$ be a convex function, and let T_n denote the composite trapezoidal approximation of $\int_a^b f(x)\,\mathrm{d}x$ with n cells. Then $T_n \geqslant \int_a^b f(x)\,\mathrm{d}x$ for all $n \geqslant 1$.

8. **Convergence estimate.** Let $f\colon [0,2] \to \mathbb{R}$ be the discontinuous function

$$f(x) = \begin{cases} 0 & \text{if } x < \sqrt{2}, \\ 1 & \text{if } x \geqslant \sqrt{2}, \end{cases}$$

and let T_n denote the composite trapezoidal approximation of $\int_0^2 f(x)\,\mathrm{d}x$ with n cells. Then there exists C such that $|\int_0^2 f(x)\,\mathrm{d}x - T_n| \leqslant C/n^2$ for all $n \geqslant 1$.

9. **Gaussian integration.** The two-point Gauss–Legendre rule on $[-1,1]$ has nodes $\pm 1/\sqrt{3}$ and both weights equal to 1.

10. **Gaussian integration.** All the weights of the n-point Gauss–Legendre rule are positive.

11. **Gaussian integration.** The n-point Gauss–Legendre rule is exact for all polynomials of degree up to $2n$.

12. **Richardson extrapolation.** Let $T(h)$ be the composite trapezoidal approximation of $\int_a^b f(x)\,\mathrm{d}x$ with step size h. Then the approximation
$$\frac{4T(h) - T(2h)}{3}$$
coincides with the composite Simpson approximation with step size h.

13. **Gaussian integration.** There exist nodes x_1, x_2 and weights w_1, w_2 such that the rule
$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx w_1 u(x_1) + w_2 u(x_2)$$
has a degree of precision equal to 3.

14. **Richardson extrapolation.** Let $f \in C^\infty[a,b]$ and let $S(h)$ denote the composite Simpson approximation of $\int_a^b f(x)\,\mathrm{d}x$ with step size h. Assuming that $S(h) = I + c_4 h^4 + c_6 h^6 + O(h^8)$, it holds that
$$\frac{16S(h) - S(2h)}{15} = I + O(h^6)\,.$$

15. **Julia.** In the following code, `R(f, a, b, n)` returns the composite Simpson approximation of $\int_a^b f(x)\,\mathrm{d}x$ with $2n$ cells:

```
function T(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    return h * (sum(f, x[2:end-1]) + (f(a) + f(b)) / 2)
end

R(f, a, b, n) = 2T(f, a, b, 2n) - T(f, a, b, n)
```

16. **Bonus.** Let $f \in C^\infty(\mathbb{R})$ and $x \in \mathbb{R}$ be such that $f''(x) \neq 0$, and let $D(h)$ denote the forward finite difference approximation of $f'(x)$:
$$D(h) = \frac{f(x+h) - f(x)}{h}\,.$$
Then the approximation
$$\frac{D(h) + D(2h)}{2}$$
satisfies $\left|\frac{D(h)+D(2h)}{2} - f'(x)\right| = O(h^2)$ as $h \to 0$. *Justify briefly.*

Answer key

Questions 1–15: 1 point for a correct answer, 0 otherwise.

1. **True.** The rule is exact for 1, x, x^2 and x^3: the right-hand sides are 1, $\frac{1}{2}$, $\frac{1}{2} - \frac{2}{12} = \frac{1}{3}$ and $\frac{1}{2} - \frac{3}{12} = \frac{1}{4}$. For x^4 it gives $\frac{1}{2} - \frac{4}{12} = \frac{1}{6} \neq \frac{1}{5}$. (This is the trapezoidal rule with the first Euler–Maclaurin correction.)

2. **False.** The rule is exact for polynomials of degree up to 3 but not for x^4: $\int_0^3 x^4 \,\mathrm{d}x = \frac{243}{5} = 48.6$, whereas the rule gives $\frac{3}{8} \cdot 132 = 49.5$. The degree of precision is 3.

3. **True.** Degree of precision at least 0 means that the rule is exact for $f = 1$, that is $\sum_{i=1}^n w_i = \int_{-1}^1 1 \,\mathrm{d}x = 2$.

4. **True.** The midpoint rule $2f(0)$ is exact for 1 and, by symmetry, for x, but not for x^2 (it gives $0 \neq \frac{2}{3}$), so its degree of precision is $d = 1$. The composite rule therefore converges with order $d + 1 = 2$: the error on each cell is $O(h^3)$, and summing over the n cells gives an error $O(h^2) = O(n^{-2})$. More precisely, the error is bounded by $\frac{(b-a)^3}{24} \frac{\max|f''|}{n^2}$.

5. **False.** The midpoint rule has degree of precision 1, so the composite rule is exact for polynomials of degree at most 1, but not for x^2. For instance, with $f(x) = x^2$ on $[0, 1]$ and $n = 1$, the rule gives $f(\frac{1}{2}) = \frac{1}{4} \neq \frac{1}{3}$.

6. **False.** The error behaves like $Ch^4 = C'n^{-4}$, so doubling n reduces it by a factor of about $2^4 = 16$.

7. **True.** On each cell, the trapezoidal rule integrates the linear interpolant of f, that is, the chord joining the endpoints of the graph. Since f is convex, this chord lies above the graph of f, so the approximation on each cell is greater than or equal to the exact integral.

8. **False.** The error estimate in C/n^2 requires $f \in C^2$, which is not the case here. Let $h = \frac{2}{n}$. Since $\sqrt{2}$ is irrational, it is never a node, and it lies in exactly one cell $[x_k, x_{k+1}]$. On all the other cells f is constant and the rule is exact. On the cell containing $\sqrt{2}$, the exact integral is $x_{k+1} - \sqrt{2} \in (0, h)$ while the rule gives $\frac{h}{2}$. The error is therefore $\left|\frac{h}{2} - (x_{k+1} - \sqrt{2})\right| \leqslant \frac{h}{2} = \frac{1}{n}$, and it is of order $\frac{1}{n}$ for many values of n: whenever $\sqrt{2}$ lies in the first or last quarter of its cell, the error is at least $\frac{h}{4} = \frac{1}{2n}$, which is not $O(n^{-2})$.

9. **True.** These are the roots of the Legendre polynomial $P_2(x) = \frac{1}{2}(3x^2 - 1)$. Exactness for 1 gives $w_1 + w_2 = 2$, and symmetry gives $w_1 = w_2 = 1$.

10. **True.** The weights are $w_i = \int_{-1}^1 \ell_i(x)^2 \,\mathrm{d}x > 0$, since $\ell_i^2 \in \mathcal{P}_{2n-2}$ is integrated exactly and $\ell_i(x_j) = \delta_{ij}$.

11. **False.** The degree of precision of the n-point Gauss–Legendre rule is exactly $2n - 1$. No n-point rule can be exact for $\prod_i (x - x_i)^2$, a polynomial of degree $2n$.

12. **True.** If $T(2h)$ is the trapezoidal rule with n cells, then $T(h)$ uses $2n$ cells of width h, and $\frac{4T(h) - T(2h)}{3}$ is exactly the composite Simpson rule on these $2n$ cells.

13. **True.** Take the two-point Gauss–Legendre rule: $x_{1,2} = \mp 1/\sqrt{3}$ and $w_1 = w_2 = 1$. It is exact for 1, x, x^3 by symmetry, and for x^2: $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$. In general, n nodes can yield a degree of precision up to $2n - 1 = 3$.

14. **True.** $16S(h) - S(2h) = 15I + c_6(16 - 64)h^6 + O(h^8)$, so the h^4 term cancels. This is the second step of Romberg integration.

15. **False.** The code uses the weights $2, -1$, which are appropriate for first-order errors, whereas the error of the trapezoidal rule is an expansion in powers of h^2. The combination `2T(f, a, b, 2n) - T(f, a, b, n)` does not cancel the h^2 term; the correct formula is `(4T(f, a, b, 2n) - T(f, a, b, n)) / 3`.

Bonus. (**False**.) By Taylor's formula,

$$f(x + h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + O(h^3),$$

and therefore

$$D(h) = f'(x) + \frac{f''(x)}{2}h + O(h^2), \qquad D(2h) = f'(x) + f''(x)h + O(h^2).$$

Consequently

$$\frac{D(h) + D(2h)}{2} = f'(x) + \frac{3}{4} f''(x)h + O(h^2),$$

which is still first order since $f''(x) \neq 0$. The correct Richardson combination is $2D(h) - D(2h) = f'(x) + O(h^2)$.